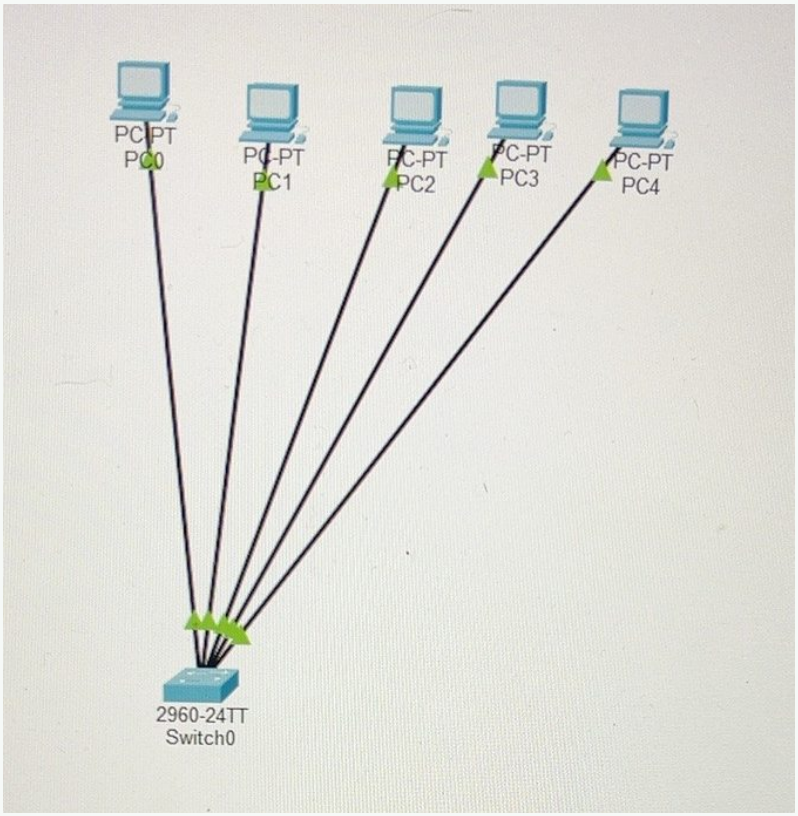
TECHNICAL CASE STUDY / 03

Cisco Packet Tracer

Five-PC Switched LAN Lab

Switch connectivity, VLAN 1 verification, IPv4 addressing and dynamic MAC learning.



SUMMER BAHAM

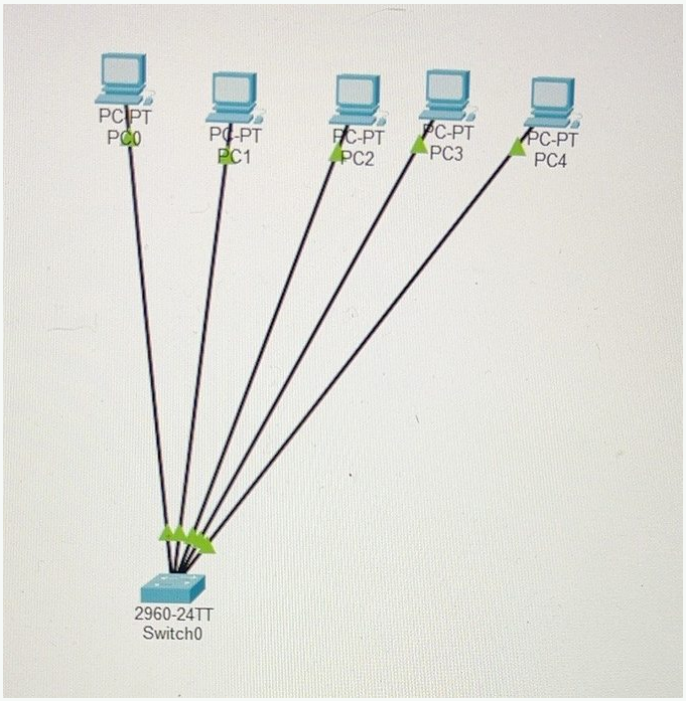
IT SUPPORT ENGINEERING / HANDS-ON PORTFOLIO

2026 • LAB REPORT

01 / PROJECT SCOPE

Topology & lab objectives

Objective: Build and verify a basic wired local area network using five Packet Tracer PCs and one Cisco Catalyst 2960-24TT Layer 2 switch.



LAB COMPONENTS

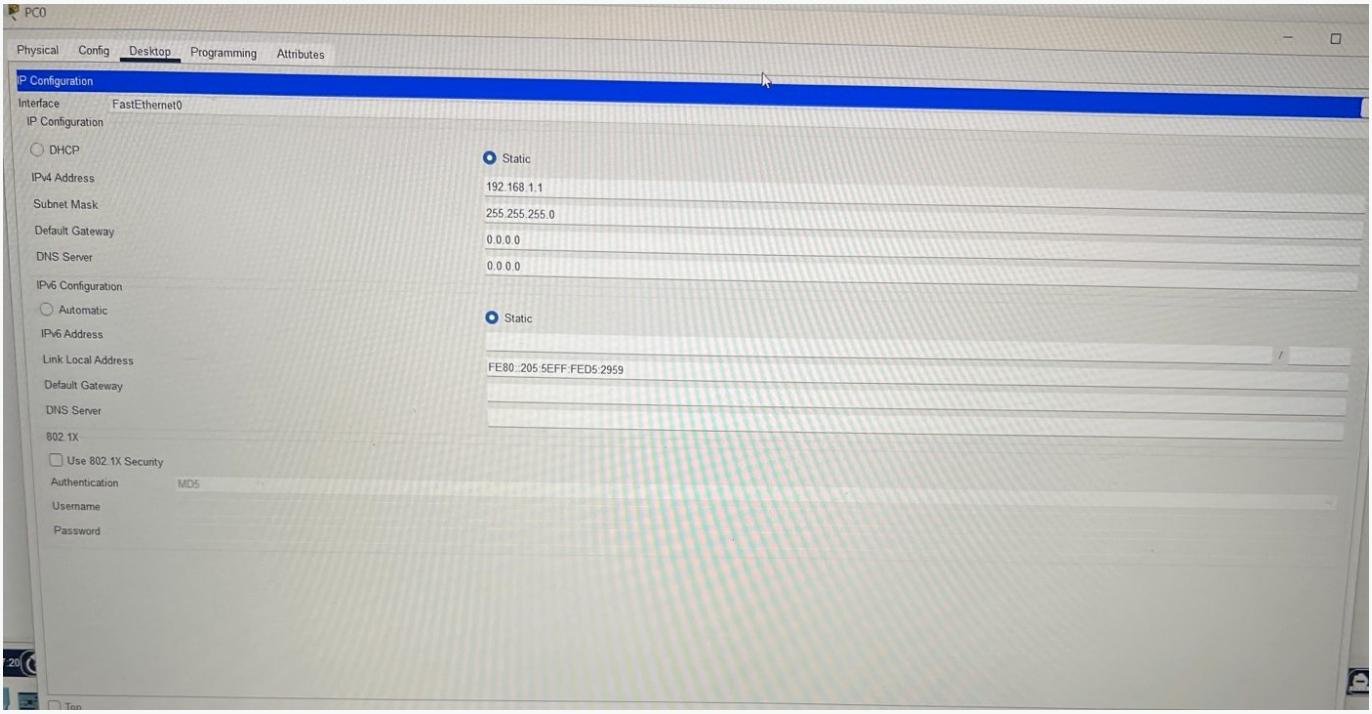
Component	Evidence from topology
Switch0	Cisco 2960-24TT
PC0 - PC4	Five attached PC-PT hosts
Connections	Five wired links; active link indicators

Scope: single switched LAN. No router, separate VLANs, DHCP service, or internet connection are shown in this scenario.

02 / HOST CONFIGURATION

PC0 static IPv4 settings

PC0 was configured using Desktop > IP Configuration. The screenshot confirms manual address assignment on the local 192.168.1.0/24 network.



CONFIRMED PC0 VALUES

Parameter	Observed setting
IPv4 address	192.168.1.1
Subnet mask	255.255.255.0
Default gateway	0.0.0.0
DNS server	0.0.0.0

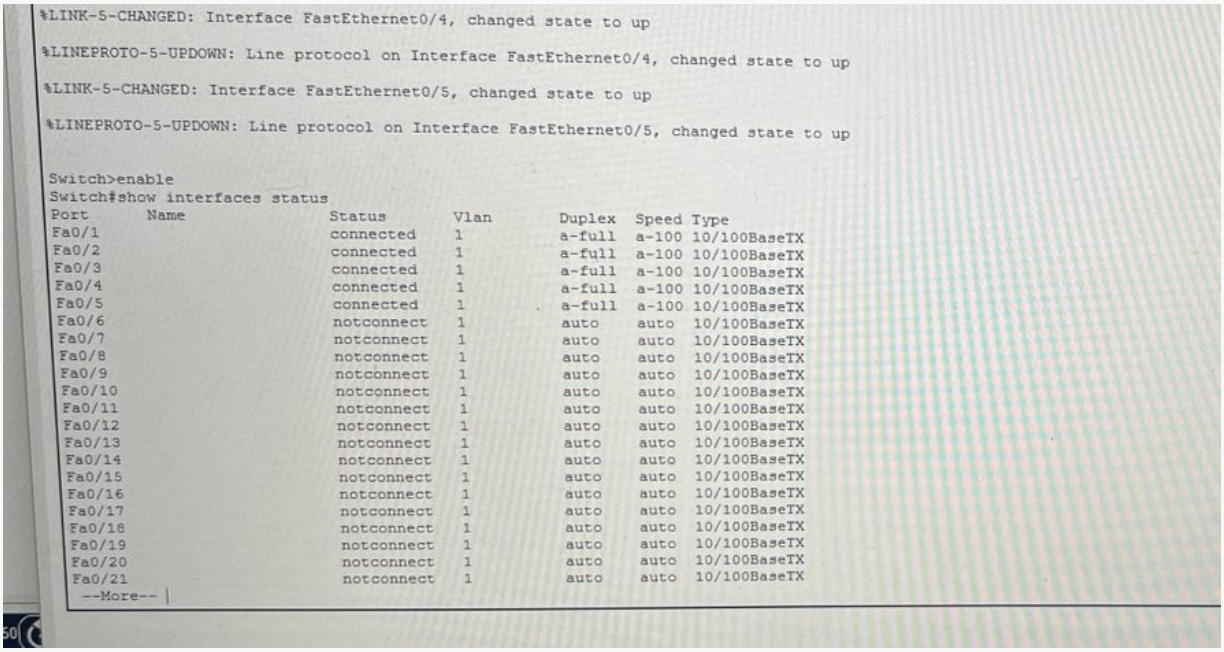
No gateway or DNS service is required to test IP connectivity between local hosts on the same subnet.

03 / SWITCH VALIDATION

Interface status & link health

The Cisco IOS command below was run at the switch privileged EXEC prompt. The output confirms five connected FastEthernet interfaces.

Switch# show interfaces status



```
%LINK-5-CHANGED: Interface FastEthernet0/4, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/4, changed state to up

%LINK-5-CHANGED: Interface FastEthernet0/5, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/5, changed state to up


Switch>enable
Switch#show interfaces status
Port      Name               Status       Vlan       Duplex  Speed Type
Fa0/1                        connected    1          a-full  a-100 10/100BaseTX
Fa0/2                        connected    1          a-full  a-100 10/100BaseTX
Fa0/3                        connected    1          a-full  a-100 10/100BaseTX
Fa0/4                        connected    1          a-full  a-100 10/100BaseTX
Fa0/5                        connected    1          a-full  a-100 10/100BaseTX
Fa0/6                        notconnect   1          auto    auto  10/100BaseTX
Fa0/7                        notconnect   1          auto    auto  10/100BaseTX
Fa0/8                        notconnect   1          auto    auto  10/100BaseTX
Fa0/9                        notconnect   1          auto    auto  10/100BaseTX
Fa0/10                       notconnect   1          auto    auto  10/100BaseTX
Fa0/11                       notconnect   1          auto    auto  10/100BaseTX
Fa0/12                       notconnect   1          auto    auto  10/100BaseTX
Fa0/13                       notconnect   1          auto    auto  10/100BaseTX
Fa0/14                       notconnect   1          auto    auto  10/100BaseTX
Fa0/15                       notconnect   1          auto    auto  10/100BaseTX
Fa0/16                       notconnect   1          auto    auto  10/100BaseTX
Fa0/17                       notconnect   1          auto    auto  10/100BaseTX
Fa0/18                       notconnect   1          auto    auto  10/100BaseTX
Fa0/19                       notconnect   1          auto    auto  10/100BaseTX
Fa0/20                       notconnect   1          auto    auto  10/100BaseTX
Fa0/21                       notconnect   1          auto    auto  10/100BaseTX
 --More--
```

VERIFIED INTERFACES

Ports	Status	VLAN	Negotiation
Fa0/1 - Fa0/5	connected	1	100 Mbps / full

Interpretation

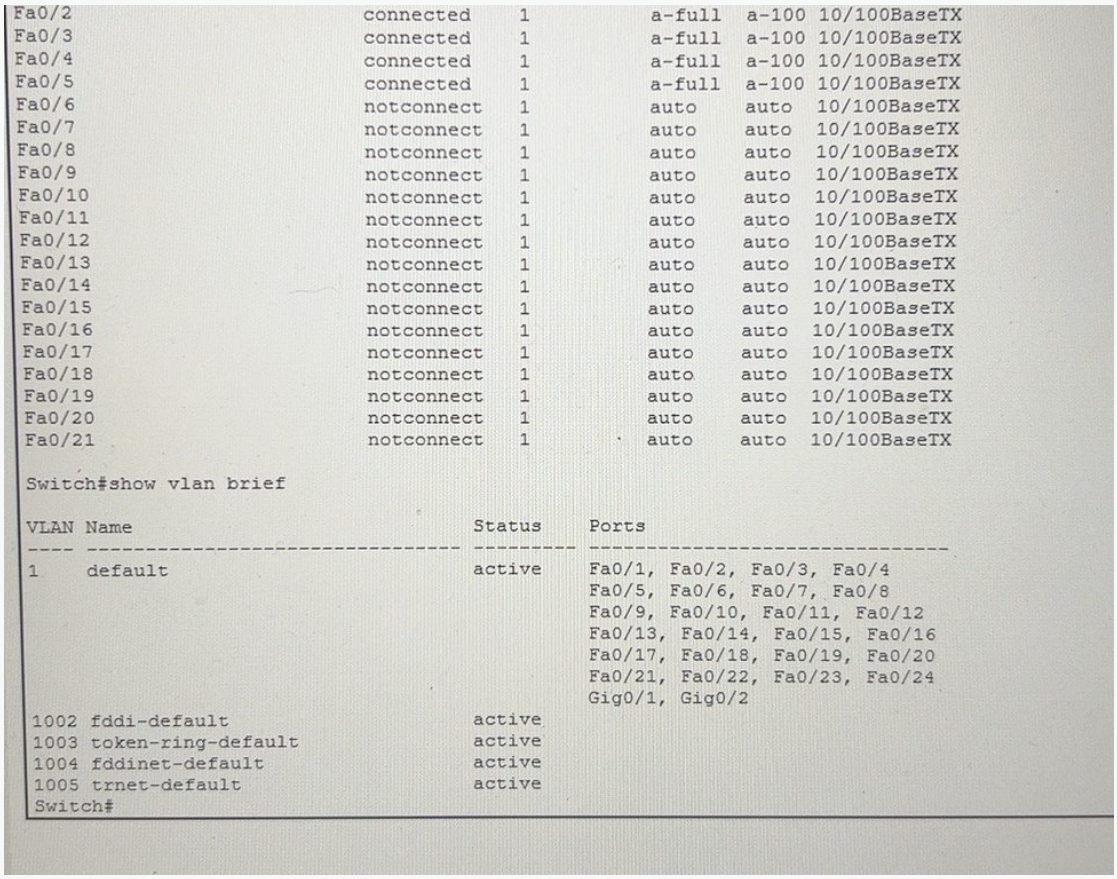
The five access links were connected at 100 Mbps, full duplex. Other displayed FastEthernet ports were not connected; this is expected for unused switch ports.

04 / LAYER 2 SEGMENTATION

VLAN membership verification

The switch was inspected with show vlan brief. All five connected access ports are members of the default VLAN 1.

Switch# show vlan brief



```
Fa0/2                     connected    1          a-full  a-100 10/100BaseTX
Fa0/3                     connected    1          a-full  a-100 10/100BaseTX
Fa0/4                     connected    1          a-full  a-100 10/100BaseTX
Fa0/5                     connected    1          a-full  a-100 10/100BaseTX
Fa0/6                     notconnect   1          auto    auto  10/100BaseTX
Fa0/7                     notconnect   1          auto    auto  10/100BaseTX
Fa0/8                     notconnect   1          auto    auto  10/100BaseTX
Fa0/9                     notconnect   1          auto    auto  10/100BaseTX
Fa0/10                    notconnect   1          auto    auto  10/100BaseTX
Fa0/11                    notconnect   1          auto    auto  10/100BaseTX
Fa0/12                    notconnect   1          auto    auto  10/100BaseTX
Fa0/13                    notconnect   1          auto    auto  10/100BaseTX
Fa0/14                    notconnect   1          auto    auto  10/100BaseTX
Fa0/15                    notconnect   1          auto    auto  10/100BaseTX
Fa0/16                    notconnect   1          auto    auto  10/100BaseTX
Fa0/17                    notconnect   1          auto    auto  10/100BaseTX
Fa0/18                    notconnect   1          auto    auto  10/100BaseTX
Fa0/19                    notconnect   1          auto    auto  10/100BaseTX
Fa0/20                    notconnect   1          auto    auto  10/100BaseTX
Fa0/21                    notconnect   1          auto    auto  10/100BaseTX

Switch#show vlan brief

VLAN Name                             Status    Ports
---- -------------------------------- --------- -------------------------------
1    default                          active    Fa0/1, Fa0/2, Fa0/3, Fa0/4
                                                Fa0/5, Fa0/6, Fa0/7, Fa0/8
                                                Fa0/9, Fa0/10, Fa0/11, Fa0/12
                                                Fa0/13, Fa0/14, Fa0/15, Fa0/16
                                                Fa0/17, Fa0/18, Fa0/19, Fa0/20
                                                Fa0/21, Fa0/22, Fa0/23, Fa0/24
                                                Gig0/1, Gig0/2
1002 fddi-default                     active
1003 token-ring-default               active
1004 fddinet-default                  active
1005 trnet-default                    active
Switch#
```

Observed VLAN state

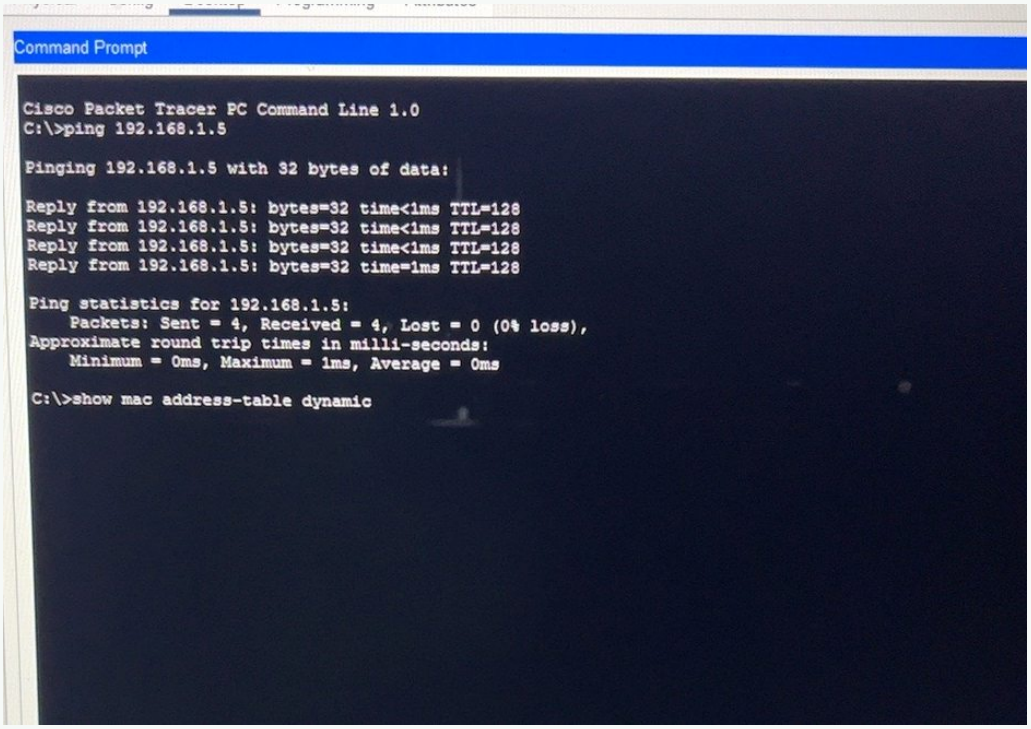
VLAN 1 (default) is active. Ports Fa0/1 through Fa0/5 are assigned to VLAN 1. The output also lists legacy default VLANs 1002-1005; no additional custom user VLANs are demonstrated.

05 / CONNECTIVITY TEST

PC0-to-target ICMP ping

From PC0, the command ping 192.168.1.5 received four responses. This verifies reachability from the documented PC0 host to the destination address.

```
C:\> ping 192.168.1.5
```



RECORDED RESULTS

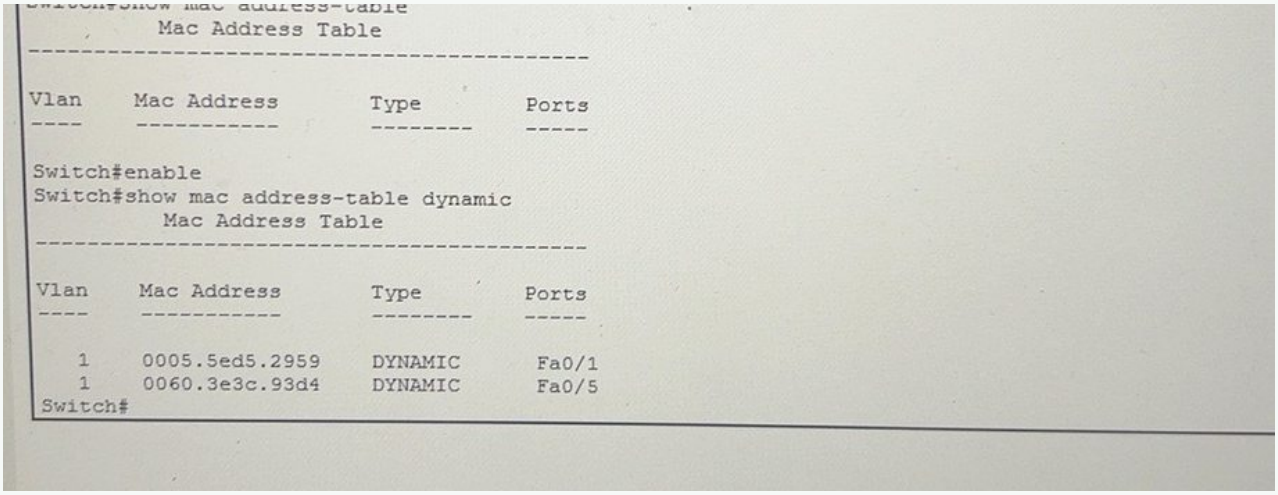
Metric	Observed result
Source PC	PC0
Destination IP	192.168.1.5
Packets sent / received	4 / 4
Packet loss	0%

06 / SWITCH LEARNING

Dynamic MAC address table

After host traffic was generated, the switch MAC address table showed two dynamic entries in VLAN 1. The earlier empty-table view was followed by evidence of successful MAC learning.

Switch# show mac address-table dynamic



```
          Mac Address Table
-------------------------------------------

Vlan    Mac Address       Type        Ports
----    -----------       --------    -----

Switch#enable
Switch#show mac address-table dynamic
          Mac Address Table
-------------------------------------------

Vlan    Mac Address       Type        Ports
----    -----------       --------    -----

   1    0005.5ed5.2959    DYNAMIC      Fa0/1
   1    0060.3e3c.93d4    DYNAMIC      Fa0/5
Switch#
```

DYNAMIC LEARNING RESULTS

VLAN	MAC address	Type	Port
1	0005.5ed5.2959	DYNAMIC	Fa0/1
1	0060.3e3c.93d4	DYNAMIC	Fa0/5

Conclusion & documentation scope

The switch learned two source MAC addresses and associated them with Fa0/1 and Fa0/5. This verifies Layer 2 address learning; screenshots do not independently establish every PC's IP and MAC address.